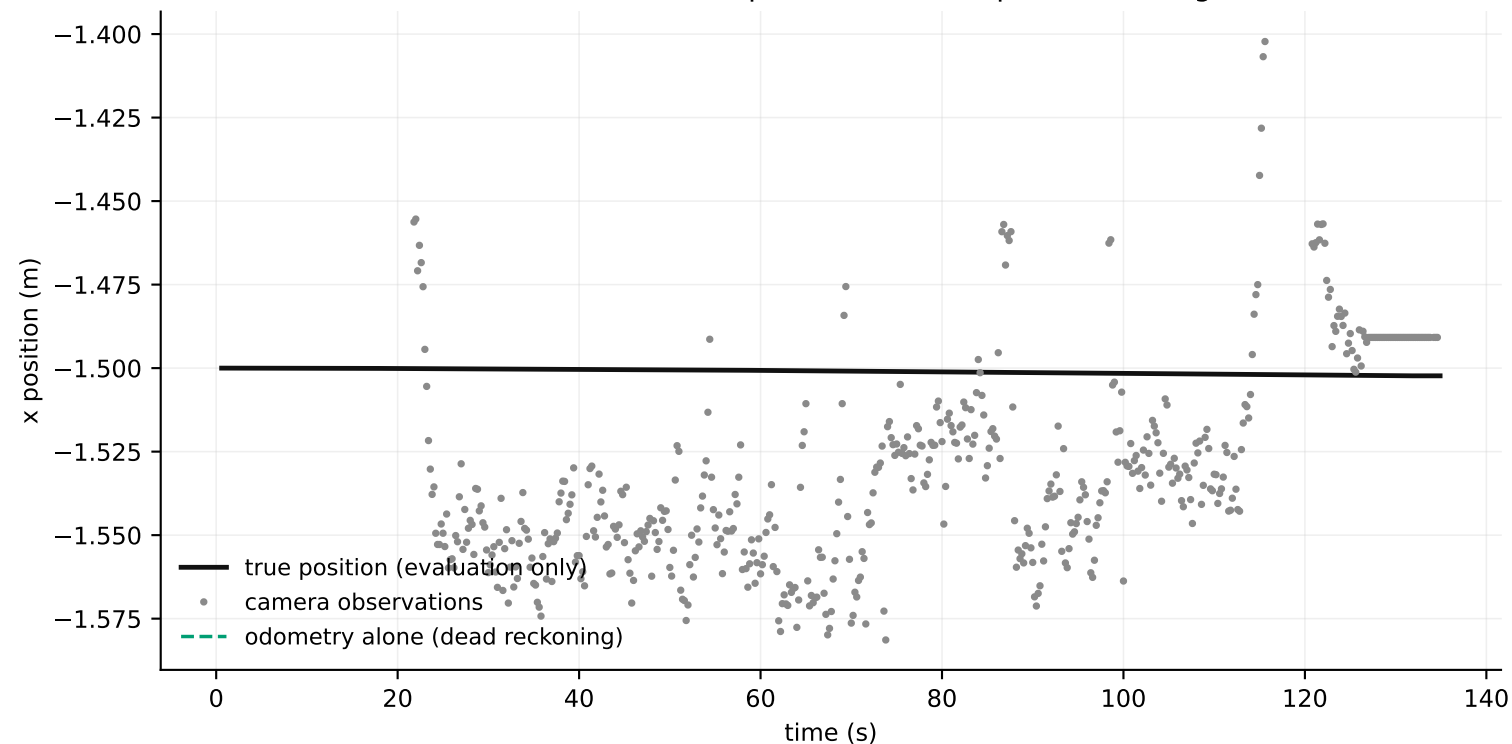


PP4 STEP 1 — write the state-space model down, then hand it the data

The data: 539 observations on 1350 steps — 60 % of steps see nothing



The generative model

$$p(z_0) = \mathcal{N}(m_0, S_0)$$

$$p(z_k | z_{k-1}) = \mathcal{N}(z_k | z_{k-1} + u_k, Q)$$

$$p(y_k | z_k) = \mathcal{N}(y_k | Cz_k, R_c)$$

$$p(y_{1:T}, z_{0:T}) = p(z_0) \prod_k p(y_k | z_k) p(z_k | z_{k-1})$$

z_k robot position (x, y)

u_k wheel-odometry increment

y_k a camera's claim about position, or missing

Q process noise — how much driving costs us

R_c measurement noise of the camera that saw it